KAIZEN MATHS

Lesson starter

Straight Line Graphs and Gradients

Finding Intercepts

Name _____

S1

Find the y-intercept of the line: y = -2x - 8

S2

Find the x-intercept of the line: y = 4x - 1

S3

Find both intercepts of the line: y = 4x - 8

S4

Find the y-intercept of the line: y = -x - 5

Show your working

KAIZEN MATHS

Lesson starter

Straight Line Graphs and Gradients

Finding Intercepts

Name _____

S1

Find the y-intercept of the line: y = -2x - 8

S2

Find the x-intercept of the line: y = 4x - 1

S3

Find both intercepts of the line: y = 4x - 8

S4

Find the y-intercept of the line: y = -x - 5

ALG-FCA4-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Straight Line Graphs and Gradients

Finding Intercepts

Name

E1

Find the x-intercept of the line: y = -3x - 7

E2

Find both intercepts of the line: y = 4x - 3

E3

Find the y-intercept of the line: y = 3x - 4

KAIZEN MATHS

Exit ticket

Straight Line Graphs and Gradients

Finding Intercepts

Name

E1

Find the x-intercept of the line: y = -3x - 7

E2

Find both intercepts of the line: y = 4x - 3

E3

Find the y-intercept of the line: y = 3x - 4

ALG-FCA4-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Straight Line Graphs and Gradients

Finding Intercepts

Name

E1

Find the x-intercept of the line: y = -3x - 7

E2

Find both intercepts of the line: y = 4x - 3

E3

Find the y-intercept of the line: y = 3x - 4

ALG-FCA4-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Straight Line Graphs and Gradients

Finding Intercepts

Name

E1

Find the x-intercept of the line: y = -3x - 7

E2

Find both intercepts of the line: y = 4x - 3

E3

Find the y-intercept of the line: y = 3x - 4

ALG-FCA4-L2 · E

Show your working

Teacher answers & working

Straight Line Graphs and Gradients · Finding Intercepts

ALG-FCA4-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Find the y-intercept of the line: y = -2x - 8

Answer:

y-intercept = -8

Working:

The y-intercept occurs when x = 0.

Substitute x = 0:

$y = -2(0) - 8$

$y = -8$

Therefore, y-intercept = -8

S2

Find the x-intercept of the line: y = 4x - 1

Answer:

x-intercept = $\frac{1}{4}$

Working:

The x-intercept occurs when y = 0.

$0 = 4x - 1$

$4x = 1$

$x = \frac{1}{4}$

Therefore, x-intercept = $\frac{1}{4}$

S3

Find both intercepts of the line: y = 4x - 8

Answer:

x-intercept = 2, y-intercept = -8

Working:

For the y-intercept, set x = 0: y = -8.

For the x-intercept, set y = 0.

$0 = 4x - 8$

$x = 2$

So x-intercept = 2 and y-intercept = -8.

Teacher answers & working

Straight Line Graphs and Gradients · Finding Intercepts

ALG-FCA4-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

Find the y-intercept of the line: y = -x - 5

Answer:

y-intercept = -5

Working:

The y-intercept occurs when x = 0.

Substitute x = 0:

$y = -1(0) - 5$

$y = -5$

Therefore, y-intercept = -5

E1

Find the x-intercept of the line: y = -3x - 7

Answer:

x-intercept = $\frac{-7}{3}$

Working:

The x-intercept occurs when y = 0.

$0 = -3x - 7$

$-3x = 7$

$x = \frac{-7}{3}$

Therefore, x-intercept = $\frac{-7}{3}$

E2

Find both intercepts of the line: y = 4x - 3

Answer:

x-intercept = $\frac{3}{4}$, y-intercept = -3

Working:

For the y-intercept, set x = 0: y = -3.

For the x-intercept, set y = 0.

$0 = 4x - 3$

$x = \frac{3}{4}$

So x-intercept = $\frac{3}{4}$ and y-intercept = -3.

Teacher answers & working

Straight Line Graphs and Gradients · Finding Intercepts

ALG-FCA4-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

Find the y-intercept of the line: y = 3x - 4

Answer:

y-intercept = -4

Working:

The y-intercept occurs when x = 0.

Substitute x = 0:

$y = 3(0) - 4$

$y = -4$

Therefore, y-intercept = -4